

Qatar-1b

R-2668

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_160928 · created 2026-08-30T01:00:44+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457659.80668 +/- 0.0009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.189 +/- 0.056
Transit depth (Rp/Rs)^2	3.56 +/- 2.12 [%]
Orbital Inclination (inc)	87.27 +/- 2.85
Airmass coefficient 1 (a1)	1.59 +/- 1.39
Airmass coefficient 2 (a2)	-0.17 +/- 0.84
Scatter in the residuals of the lightcurve fit is	103.14 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	5.75
Transit Duration (day)	0.0803 +/- 0.009

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.189 $\pm$ 0.056 vs 0.1463 (deviation 0.76 σ)
Duration fit vs published	0.0803 d vs 0.0692 d (deviation 1.24 σ)
Mid-time O-C	-1.4 min
Depth SNR	3.4
Residual scatter	103.14 %

Light curve

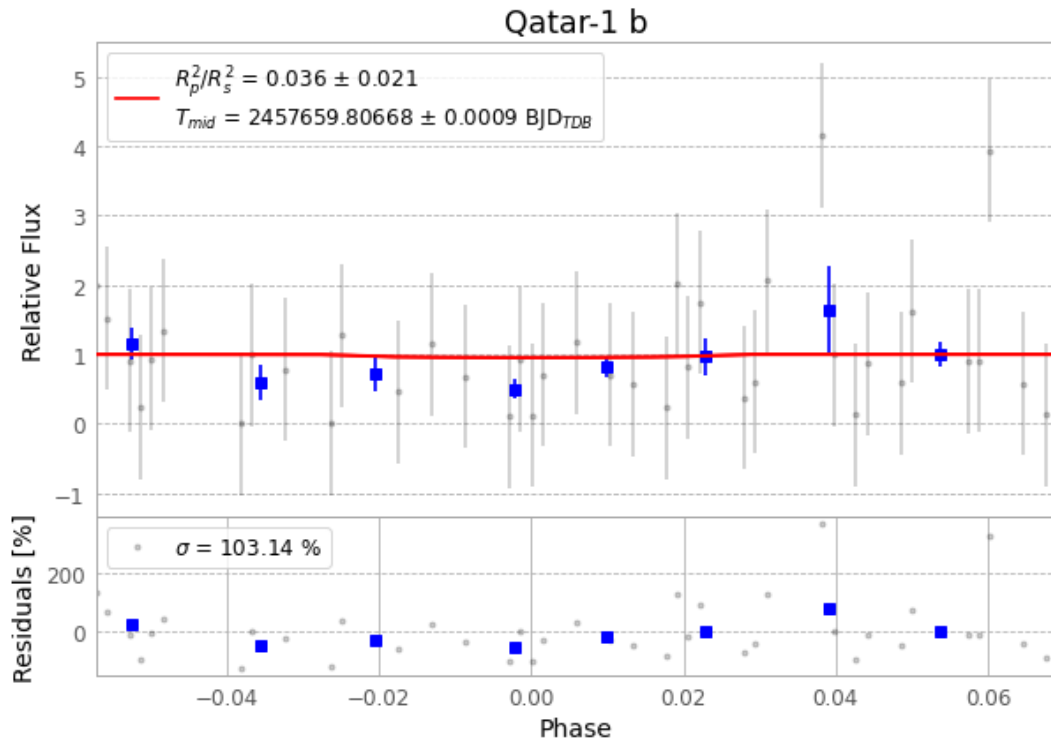


Figure 1 — FinalLightCurve_Qatar-1 b_2016-09-27.png (SHA-256 7ffddea53ad151a4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [460, 184], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 79bac201721b8bc2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

98 FITS frames (64 MB), Qatar-1160928052112.FITS → Qatar-1160928101212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-160928.log (c58daa495e87...), FinalLightCurve_Qatar-1 b_2016-09-27.png (7ffddea53ad1...), FinalLightCurve_Qatar-1 b_2016-09-27.csv (d98ae7e345c5...)

Compute resources

Wall time 148.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 01:00Z.